

Prince Acquah Rockson

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Profile

I am a Biomedical Data Science/AI Researcher with experience in both wet lab and computational skills. My recent research has been in building robust and interpretable machine learning systems for healthcare applications, including chronic kidney disease prediction and staging. Over time, I have developed an interest in leveraging computational approaches, including probabilistic machine learning, causal inference, and adaptive decision-making, alongside wet-lab skills to early detect and manage non-communicable diseases.

Education

Kwame Nkrumah University of Science and Technology

HEALTH INFORMATICS, Master of Science

Kumasi, Ghana

Jan. 2024 - Nov. 2025

University of Cape Coast

MOLECULAR BIOLOGY AND BIOTECHNOLOGY, Bachelor of Science

Cape Coast, Ghana

Sept. 2014 - May 2018

Research Experience

- **Topic:** Development and Validation of an Explainable Framework for 6-Class Chronic Kidney Disease Staging: A Computational Diagnostic Study
 - **Status:** Manuscript submitted to BMC Medical Informatics and Decision Making (Awaiting Preprint)
 - **Key Contribution:** Engineered a multi-stage diagnostic framework using ML algorithms such as Gradient Boosting Machine, LightGBM, XGBoost, and others. Implemented 100-iteration bootstrapping for stability and SHAP for clinical interpretability.
 - **Role:** Principal Researcher / Lead ML Architect.
 - **Advisor:** Dr. Eric Osei Opoku
- **Topic:** Detection Of Genetically Modified Products Using PCR (2018)
 - **Focus:** Biotechnology / Molecular Biology
 - **Key Contribution:** Performed laboratory-scale genomic analysis; established the wet-lab foundation that informs my current understanding of molecular health data.
 - **Role:** Principal Researcher .
 - **Advisor:** Dr Emmanuel Plas Otwe

Professional Experience

Halges Financial Technologies (Korba)

Accra, Ghana

Data Scientist (BI, Audit & Transaction Intelligence Functions)

June 2024 - Present

- Engineered a Python and Streamlit SLA audit platform that automates pricing compliance verification across multi-segment transactions; reduced manual validation overhead by 60% while eliminating revenue leakage.
- Designed automated data quality pipelines and statistical validation checks across SQL/Python environments, identifying transactional discrepancies and strengthening database integrity across high-volume banking, wallet, and merchant ledgers.
- Developed an automated Python ingestion engine that processes raw transactional data, executing dynamic schema mapping, business-logic segmentation, and statistical pivot aggregations with interactive Plotly data visualizations.
- Implemented a rule-based Machine Learning/Python decision engine to classify and rank Change Advisory Board (CAB) requests based on multi-factor risk assessments and operational urgency metrics.
- Built scalable dashboard solutions featuring dynamic column mapping, automated error correction, and multi-sheet automated reporting, replacing legacy Excel workflows and significantly reducing time-to-insight.
- Partnered cross-functionally with Product and Finance teams to translate complex transactional datasets into executive-level KPIs, driving operational efficiency and supporting strategic decision-making.

Docupro Ltd

Accra, Ghana

Lead Health Data Quality Analyst,

May 2020 - June 2024

- Engineered automated Python and SQL validation scripts to audit large-scale, heterogeneous national health datasets, systematically detecting and resolving missingness, duplications, and structural anomalies.
- Monitored and optimized enterprise ETL data pipelines, ensuring strict adherence to data governance protocols for completeness, schema consistency, latency reduction, and downstream analytical readiness.
- Directed weekly quality control protocols across distributed national healthcare databases, establishing standardized data profiling techniques to preserve signal integrity for clinical and operational analytics.
- Partnered with data engineers and health domain experts to redesign data integration workflows, enhancing systemic validation logic and enterprise-level data governance.
- Authored comprehensive national health data quality reports, delivering statistical insights and algorithmic process improvements to senior leadership to strengthen decision support systems.

S.K. Consult

Cape Coast, Ghana

Associate Consultant

Mar 2020 - Dec. 2023

- Collaborated with at least 100 clients to define their research topics, objectives, and scope.
- Collected quantitative and qualitative data through surveys and interviews for at least 100 clients.
- Developed and selected appropriate research plans, research methods, and data analysis tools.
- Analysed and interpreted both primary and secondary quantitative/qualitative data using statistical tools like SPSS, R, Stata, PLS-SEM, and analytical techniques like t-tests, MANOVA, and ANCOVA.
- Generated reports and presented findings with appropriate visual representations

Parockson Digital Solutions

Accra, Ghana

Head of Projects

March 2019 - Present

- Leading end-to-end planning and execution of software and AI research projects.
- Supervising cross-functional teams to deliver client solutions on schedule.
- Managing project budgets, timelines, and stakeholder communications.
- Oversaw the design and rollout of digital solutions for clients.
- Ensured quality assurance and documentation for all completed projects.

Biotechnology and Nuclear Agriculture Research Institute

Accra, Ghana

Research Assistant and Overseer of BSF lab (NSS and Volunteer)

Sep. 2018 - May 2020

- Designed and executed controlled bio-conversion experiments on Black Soldier Fly (BSF) growth optimization; applied biostatistical techniques to model biomass accumulation and organic waste conversion rates.
- Cleaned, structured, and analyzed complex experimental datasets using quantitative research methodologies, translating raw biological metrics into statistical trends and actionable scientific insights.
- Partnered on cross-functional biotechnology projects, contributing to experimental setup, statistical hypothesis testing, and quantitative modeling for sustainable agricultural systems.
- Directed daily BSF laboratory workflows and supervised junior research assistants, enforcing data integrity standards, rigorous laboratory protocols, and experimental quality control.
- Co-authored technical reports and scientific manuscript drafts, transforming complex biological findings and statistical outputs into executive summaries and publication-ready documentation.

Daasgift Quality Foundation

Takoradi, Ghana

Monitoring & Evaluation (M&E) Intern

June 2017 - Aug. 2017

- Collected and analysed project data to track performance indicators.
- Assisted in preparing weekly M&E reports and documentation for donor submissions.
- Supported field data collection and beneficiary feedback sessions.
- Helped design survey tools and evaluation templates for ongoing projects.
- Contributed to data entry, cleaning, and visualisation for impact assessments.

Justab Hospital

Accra, Ghana

Medical Lab Intern

June 2016 - Aug. 2016

- Assisted with specimen reception, preparation, and routine laboratory workflow under supervision.
- Supported basic diagnostic procedures, reagent preparation, and equipment cleaning in line with laboratory safety standards.

- Maintained accurate records and observed quality control practices in medical laboratory operations.

Justab Hospital

Accra, Ghana

Medical Lab Intern

June 2015 - Aug. 2015

- Supported laboratory sample handling, preparation, and documentation for routine clinical investigations.
- Observed and assisted with basic haematology, microbiology, and clinical chemistry workflows under supervision.
- Practised laboratory safety, infection prevention, and proper handling of biological specimens.

Deogratias Institute

Kasoa, Ghana

Teacher

July 2013 - Sep. 2014

- Designed and administered standardized evaluation frameworks to measure learner progress; analyzed student academic performance metrics to identify learning gaps and optimize instructional delivery.
- Maintained structured data logging systems for student attendance, grade tracking, and performance records, ensuring accurate record-keeping and data accessibility.
- Developed structured lesson plans and technical learning modules, translating complex subject matter into clear, digestible concepts for diverse student audiences.
- Mentored students in academic development and critical thinking; facilitated collaborative problem-solving and maintained structured communication with stakeholders to support learning outcomes..

Peer Review Profiling

- ACM RESPECT 2025 Conference (Association for Computing Machinery)
 - Evaluating research submissions focused on equity and computer science education research.

Workshops, Webinars & Conferences

Faculty of Physical and Computational Sciences PosterFest | Aug 2025 | Kumasi, Ghana

- Presenter: "Detection and Staging of Chronic Kidney Disease: A Step Toward Preventive Nephrology."
- Highlighted the integration of Machine Learning for subclinical diagnosis in resource-limited settings.

Centre for Business Analysis (Virtual) | Aug 2025

- Invited Speaker: "The Future of Data Analytics: How Certifications Keep Us Relevant."
- Discussed the evolving landscape of AI and the necessity of continuous skill acquisition in data science.

ASPIRE 2025 Program (Virtual) | Aug 2025

- Keynote Speaker: "Telling the Data Story."
- Trained emerging professionals on data visualization and communicating complex insights to non-technical stakeholders.

Bayobab Africa (Virtual) | Aug - Nov 2025

- Technical Speaker: "ITIL Frameworks and Cloud Computing Service Life Cycles."
- Delivered a modular series on service improvement, stakeholder management, and cloud infrastructure roles.

Leadership Roles & Experiences

Change Advisory Board Member

Accra, Ghana

Halges Financial Technologies Ltd

Mar., 2026.- Present

- Risk Assessment: Evaluating how a proposed update might affect existing workflows or data integrity
- Stakeholder Alignment: Ensuring that changes to communication channels or branding assets maintain the established visual identity and professional reputation.
- Schedule Coordination: Reviewing the "Change Calendar" to ensure updates do not conflict with major events.
- Post-Implementation Review (PIR): Analyzing whether a change achieved its desired outcome and documenting lessons learned to improve future deployment cycles.

User Acceptance Testing (UAT) Team Member

Accra, Ghana

Halges Financial Technologies Ltd

, July., 2024.- Present

- Test Case Development: Creating detailed scenarios based on user requirements.

- User Feedback Synthesis: Managing pilot groups of alumni to gather qualitative data on UI/UX and translating that into actionable bug reports for developers.
- Defect Management: Using tools like GitHub issues to track functional gaps, ensuring the final product reflects the intended branding and professional standards.
- Final Sign-off: Providing the formal "Go/No-Go" recommendation before a platform or tool is launched to the wider network.

Team Lead

Accra, Ghana

Jospong ICT Data Science Team for Shared Services

Jan. 2024- Present

- Led a cross-functional data science team delivering analytics and automation solutions for shared services units
- Oversaw end-to-end project execution, from problem definition and data collection to model development, validation, and deployment
- Guided junior analysts through technical reviews, mentoring, and skill development
- Trained and mentored at least 15 data analysts and scientists for shared services.

Skills

Computational Research & Data Science

- **Languages:** Python (Advanced: NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn), R (Statistical Modelling), SQL (PostgreSQL, MySQL).
- **Statistical Analysis:** SPSS, Stata, PLS-SEM, Excel.
- **Specialised AI:** Machine Learning Model Validation (Bootstrapping, Cross-Validation), Explainable AI (SHAP/LIME).
- **Data Visualisation & Business Intelligence:** Power BI, Advanced Excel (VBA/Macros), Tableau.

Software Engineering & Deployment

- **Web & App:** React JS/TS, Node.js, Streamlit (for ML Dashboarding).
- **DevOps & Infrastructure:** Git/GitHub, Netlify, Render, Railway, Vercel.
- **IT Architecture:** ITIL Frameworks, Cloud Computing Service Lifecycle.

Wet-Lab, Molecular Biology & Biotechnology

- **Laboratory Techniques:** PCR, gel electrophoresis, DNA extraction, sample preparation, microscopy, aseptic technique, media preparation, serial dilution, and basic microbiological culture handling.
- **Molecular Biology & Biotechnology:** genomic analysis, molecular diagnostics support, biotechnology research protocols, biological sample handling, and experimental documentation.
- **Medical Laboratory Practice:** Laboratory safety, specimen reception and processing, quality control, reagent preparation, equipment maintenance, and accurate laboratory record keeping.

Training and Certifications

Centre For Data Science and Analytics/ Ashoka University

Virtual

Data Analytics For Climate and Health

Sep. 2025

Azubi Africa

Virtual

Data Analytics Professional

Aug. 2024

Coursera

Virtual

Google Data Analytics; Google Project Management; Google IT Support

Jun 2023- May 2024

Amalitech

Virtual

Software Development Programme

Sep. 2023

Centre For Business Analysis

Accra, Ghana

Certified Business Data Analyst

Aug. 2022

Google Africa Developer Training/ Andela

Virtual

Android Development

May. 2022

Languages

- **English:** Native / Professional Fluency (Full Research Proficiency).
- **Local Languages:** Asante Twi, Fante, Ga.